

1 Introduction

During this sprint, we continued working hard. We continued improving existing features while also adding new ones. We improved our edge detection algorithm, cornering and parking, expanded our path planning, fixed hardware issues, mounted additional sensors. We also implemented object avoidance and overtaking with the help of those additional sensors. As for our previous pedestrian detection issue, we decided to redo our entire object detection dataset and have a single model looking for objects on the entire camera feed. This sprint, we encountered minor issues, mostly hardware-related. Mainly, our steering servo's arm was in bad shape since we received the car, and this sprint it finally broke. After a surprising amount of searching we managed to find a replacement and keep working.

2 Planned activities

During this sprint, we had a plan to accomplish the following tasks:

- Optimize where possible
- Traffic light and pedestrian reaction
- Handle overtaking
- Place additional sensors on platform
- Handle obstacle avoidance
- Path replanning

3 Status of planned activities

- **Optimize where possible**

Our biggest optimization this sprint was completely redoing our entire object detection dataset from the ground up, using the native resolution that we get from the camera. This, when paired with running just one model gives us a relatively fast and robust image processing algorithm. Additionally, doing this also fixed an issue we discovered with detecting traffic lights. We also noticed the speed calibration was way off, so we fixed it. Our edge detection algorithm is now also more robust but we aren't satisfied with the current throughput of ~10 images per second, so we intend to optimize it further in the next sprint.

- **Traffic light and pedestrian reaction**

While we were in the process of implementing our two model idea from the last sprint, we noticed an issue where traffic lights would only be detected from far away. This prompted us to redo the entire dataset in the native resolution from the camera. Although time-consuming, this change improved efficiency, allowing us to merge the two models into one and have the algorithm process frames directly from the camera without resizing, which ultimately speeds everything up.

- **Handle overtaking**

When the car detects a moving obstacle, it moves to the left lane and speeds up until it passes the obstacle. This feature has not been fine-tuned, and we are not yet satisfied enough to demonstrate it in action.

- **Place additional sensors on platform**

We placed our sensors and microcontroller near the front, where it would make most sense for them to go. We didn't want to make any permanent modifications to the car shell yet, so for now everything is securely held on by tape and zip-ties.



- **Handle obstacle avoidance**

When the car detects an obstacle that isn't moving, it stops for a second and then moves past it, coming back to its lane before continuing on its way.

- **Path replanning**

We haven't gotten to this task yet as we deemed it a non-urgent issue.

4 General status of the project

Since the last report, the car has been improved nearly every way. As for hardware, we replaced our failing servo arm, secured all of our components into a nice, tight fit, and added a polarizing film over the camera's lens to reduce glare. We've improved the robustness of our edge detection algorithm, however we aren't satisfied with its speed and we intend to fix that next sprint. We've overhauled our object detection algorithm, added overtaking and object avoidance, calibrated the speed and steering, fixed how the car handles turns in intersections, integrated our parking sensors with the actual parking script. Now the car can park on either side of the road, wherever the nearest parking spot is. We've started development on handling intersections, however we aren't satisfied with our results enough to show them yet and are working on a fix. We made a physical roundabout for the car to drive on and have begun testing and correcting how it reacts to it.

5 Upcoming activities

In the next sprint we want to do the following tasks:

- Optimize current codebase wherever possible
- Mechanical maintenance
- Finish roundabout handling and overtaking
- Integrate localization for path planning/replanning